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Transcriptional coupling of neuronal fate commitment and the onset of migration

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During mammalian CNS development, when the neural precursor cells commit to the neuronal fate they must delaminate and migrate toward the pial surface in order to reach the appropriate final location. Thus, the coordination of delamination and fate commitment is important in creating the correct structure. Although previous studies have proposed that spindle orientation during mitosis plays a role in both delamination and fate commitment, thus coordinating these events, subsequent studies have challenged this model. Recent work has identified several transcriptional mechanisms associated with neurogenesis that inhibit cell adhesion of newborn neurons and intermediate neuronal progenitors, thereby triggering delamination and linking it with fate commitment.

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Introduction

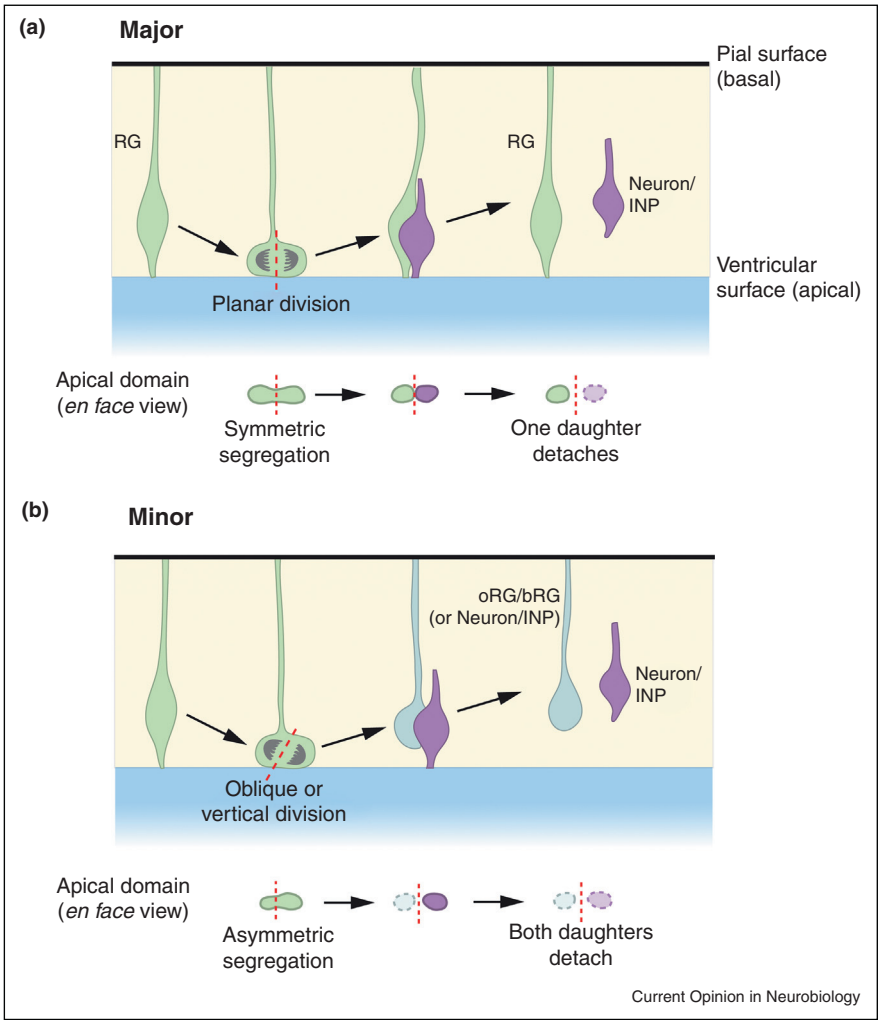
Epithelial cells, including the neuroepithelium, have apical-basal polarity, with apical adherens junctions between neighboring cells helping to maintain this polarity [1,2]. The cell bodies of neuroepithelial cells are located within the ventricular zone (VZ) with the apical end of the cells facing out into the ventricle and the basal end, a long thin process, extending to the pial surface. These cells, called neural precursor/progenitor cells (NPCs), divide symmetrically in the early stage to expand the progenitor pool, and then after that can divide either symmetrically or asymmetrically, at which point they are also known as radial glial cells (RGs). Symmetric divisions produce two RGs, which remain at the ventricular surface. Asymmetric divisions can result in one RG, a neuron that detaches from the surrounding cells and migrates basally, or an intermediate neuronal progenitor (INP) that also migrates basally and then divides at least once to make two or more neurons [1,2]. Newborn INPs and neurons express proneural genes, including Neurogenins and Ascl1 [3,4], which are thought to confer the neuronal fate. After neuronal fate commitment, cells migrate toward the pia and settle outside of the other neurons [5–8], meaning that a coupling between the onset of migration and fate commitment is important in making sure that the neurons of each subtype settle in the appropriate places. Here we present various mechanisms that initiate the first steps of migration and discuss how these events might be coordinated with neuronal fate commitment.

Spindle orientation and the onset of neuronal migration

Several studies have suggested that spindle orientation during cell division plays a role in determining both the neuronal fate and onset of migration of the daughter cells [1,9], proposing one mechanism of coordination between the onset of migration and fate commitment. These models propose that if the spindle is oriented parallel to the apical surface, it will result in a planar division in which the apical surface is divided evenly between the two daughter cells, leading to both remaining in the neuroepithelium and both becoming RGs. Conversely, if the spindle is oriented perpendicular to the apical surface (vertical), or in another such way that causes this domain to be divided unevenly between the daughters (oblique), the cell that inherits more of the apical surface will become a RG and the other will migrate basally and thus become an INP or neuron. Proponents of this type of model propose that the apical surface includes fate determinants and that spindle orientation influences the amount of apical surface inherited by each cell, which then determines the cell fate and whether the cell stays at or migrates away from the apical surface [10–13].

However, in the last several years, studies have revealed that, under normal conditions, spindle orientation is likely not the main determinant of whether cells remain at the apical surface (and become RGs) or migrate away (and become INPs or neurons) [14–16,17•,18]. Analysis of spindle orientation during division in both the rodent cortex and the chick spinal cord has revealed that most divisions are planar [14–16,19•], and, furthermore, time lapse imaging of the apical surface, labeled with eGFP-tagged junction protein ZO1, showed that it is almost always divided symmetrically between the two daughter cells [15,17•,20] (Figure 1). Furthermore, another study

Figure 1



Division of RGs. **(a)** The majority of RGs (green) undergo planar divisions in which the apical domain is segregated symmetrically between both daughters. One of the daughters (green) maintains both apical and basal attachment and self-renews, while the other (purple) eventually detaches from the ventricular surface and becomes a neuron or INP. **(b)** The minority of RGs (green) undergo oblique or vertical divisions in which the apical domain is segregated asymmetrically giving rise to one cell that immediately detaches from the VZ but inherits the basal process (blue) and another that initially remains attached to the VZ (purple). The former is thought to become an oRG/bRG (or a neuron or INP), whereas the latter eventually detaches from the ventricular surface and becomes a neuron or INP.

that monitored both the apical surface segregation and whether the cells delaminated after division found that there was no difference in the size of the apical surface inherited between cells that delaminated and those that did not [17•]. Because only a small proportion RGs divide with vertical or oblique spindle orientation and since the daughter cells can still delaminate even if they inherit the apical surface (Figure 1), it is unlikely that spindle orientation, and the resultant symmetric or asymmetric apical surface inheritance, is the main determining factor in the onset of migration.

That being said, evidence from experiments that randomize spindle orientation lend support to the idea that spindle orientation can affect delamination (and cell fate), at least in some respects [21]. Upon deletion [15] or expression of a dominant negative form of LGN [14,17•,22], which is important for orienting the spindle parallel to the apical surface, a larger proportion of cell divisions were vertical or oblique, and a higher number of NPCs were found outside of the ventricular zone. This suggests that ectopic vertical spindle orientation is able to initiate migration. Furthermore, overexpression (or knock in) of mouse Inscuteable, which helps to orient the spindle perpendicular to the apical surface, resulted in even more vertical or oblique divisions than in the LGN deletion and also increased the number of INPs [15,19•]. Although spindle orientation can clearly influence the

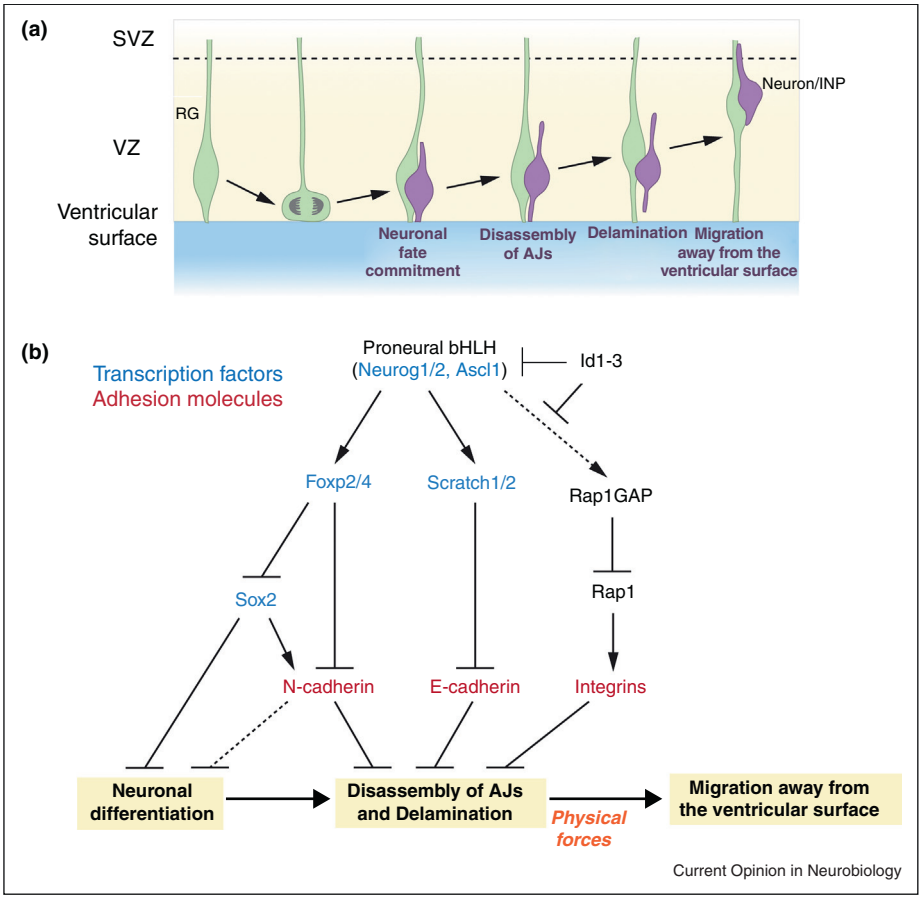
organization of the developing brain, the fact that vertical or oblique orientations are rare in the developing cortex together with the finding that decreasing the number of planar orientation divisions did not decrease the number of progenitors [14,15] suggests that spindle orientation is not the main factor that regulates the onset of migration or its coupling with neuronal fate commitment. Instead, it appears that under wild type conditions, most RG divisions have planar spindle orientation and both cells initially remain at the apical surface, leaving the question of what coordinates cell fate determination and the onset of migration in these seemingly symmetrical divisions.

Delamination and subsequent initial movement of newborn neurons and INPs

The onset of migration has two steps (Figure 2a): delamination, which involves dissolution of the connections that cells have at the apical surface, that is, adherens junctions (AJs), and the initial movement out of the VZ, which is mediated by physical forces.

These physical forces include both intracellular forces from cytoskeleton reorganization as well as extracellular forces that push newborn neurons/INPs away from the apical surface. Early support for the role of intracellular forces comes from identification of FilaminA mutations in patients suffering from periventricular nodular heterotopia, a congenital malformation of the cerebral cortex characterized by ectopic neuronal clusters along the ventricle, which could result from defective migration [23,24]. Since FilaminA regulates actin-cross-linking and is known to promote migration in other tissues, it was proposed that actin reorganization could also generate intracellular forces that support migration of neurons and INPs. Interestingly, Ngn2 and Asc1 were recently shown to activate Rho GTPases Rnd2 and Rnd3, respectively [25,26•]. Rnd2 and Rnd3 induce actin reorganization through inhibition of RhoA activity to generate forces to activate motility [26•], suggesting one possible way that departure from the VZ might be coordinated with fate commitment. It is possible that these molecules also regulate delamination, which we

Figure 2



The onset of neuronal migration. **(a)** Following a planar RG division, one daughter cell has committed the neuronal fate (purple) while one remains an RG (green). Upon neuronal fate commitment, the newborn neuron or INP dissolves its AJs, delaminates from the ventricular surface and then migrates out of the VZ. **(b)** Transcriptional regulation of the disassembly of AJs and delamination of newborn neurons and INPs.

will discuss later. In addition to such intracellular cytoskeletal forces, the cell bodies and nuclei of VZ cells seem to receive external passive forces in a direction away from the apical surface. RG cell nuclei undergo interkinetic movement periodically during the cell cycle. A recent paper proposed that active accumulation of M-phase cells at the apical end causes 'a collective displacement effect' to make other cells passively move away from cell-dense regions [27]. The loss of apical attachments may make newborn neurons and INPs more susceptible to such external forces, promoting their movement out of the ventricular zone.

In order for these physical forces to successfully activate migration, the newborn neurons and INPs must first attenuate their intercellular adhesions at the apical surface, that is, AJs, meaning that AJ dissolution is an important step for the onset of migration and potentially the development of brain and spinal cord structure. In fact, inactivation of molecules that are associated with adhesion machinery, such as E-cadherin or N-cadherin, causes disintegration of AJs and brain malformation, whereas forced expression of E-cadherin or N-cadherin can prolong the maintenance of radial morphology and apical attachment of RGs [28–31]. Thus, AJ-mediated adhesion appears to regulate the onset of delamination of newborn neurons and INPs.

Transcriptional mechanisms coordinating neuronal fate commitment and delamination

Since coordination between fate commitment and delamination is likely important for the appropriate positioning of neurons, it would be logical if proneural genes both promoted neuronal fate commitment and played a role in activating the dissolution of AJs that marks the first step in migration. In fact, Neurogenins and Ascl1 have been shown to play central roles in coupling neurogenesis and the physical movement of newborn neurons out of the ventricular zone [25,26•,32,33], and it is likely that they also regulate delamination. As described above, these genes activate Rnd2 and Rnd3, which contribute to actin reorganization via RhoA inhibition [25,26•]. Given that RGs form dense junctional actin bundles at the apical surface in a RhoA-dependent manner [34–37], it is conceivable that Rnd2 and Rnd3 might also be involved in the delamination process through disintegration of AJs. However, a recent study suggests that Rnd3 maintains AJ integrity [38], supporting the idea that Rnd proteins may not participate in the disassembly of AJs. Instead, recent reports have uncovered novel mediators that interact with neurogenic pathways, thus affecting neuronal fate commitment, while also functioning to regulate detachment.

Among such mediators are Id proteins, which antagonize the bHLH class of transcription factors, including Neurogenins and Ascl1, and have been shown to play essential roles in maintaining the undifferentiated state of RGs and in preventing detachment from the apical surface. In the developing cortex, deletion of Id1–Id3 caused RGs to detach from the ventricular surface and prematurely differentiate [39•]. This early detachment is due to Id proteins' regulation of the GTPase activity of RAP1, a key mediator of cell adhesion [40], via repression of *Rap1GAP*, which converts RAP1 to its inactive (GDP-bound) form [39•]. In fact, Id repression of *Rap1GAP* is required to maintain NPCs/RGs' adhesion to extracellular matrix proteins fibronectin and laminin *in vitro* and their adhesion to the apical surface in the embryonic brains [39•,41] (Figure 2b). Since Id proteins are highly expressed in the VZ in the developing cortex, it is conceivable that RGs have high RAP1 activity, promoting cell adhesion and residence at the VZ, while newly fate committed neurons or INPs exhibit lower RAP1 activity. These data illustrate one potential mechanism by which cell fate determination and delamination are coordinated: suppression of Id proteins activates proneural bHLH factors, promoting the neuronal cell fate and also regulating the cell adhesion mediator, RAP1, to induce detachment.

Recent findings have also reported that N-cadherin downregulation, which could induce delamination, might be regulated by factors that also promote neuron or INP production, thus coordinating neuronal differentiation and delamination. During motor neuron development in the chick spinal cord, Forkhead transcription factors, Foxp2 and Foxp4, downregulate N-cadherin to induce delamination as well as Sox2 to promote neuronal differentiation [42•] (Figure 2b). While Foxp proteins were shown to directly downregulate N-cadherin, it is unclear whether this direct repression is the main mechanism by which they inhibit N-cadherin expression or if they also regulate its expression through their downregualtion of Sox2. While Foxp2 was expressed at similar levels both in precursor and newly differentiated cells, Foxp4 expression was elevated ~2-fold upon neuronal differentiation, suggesting that increased Foxp2/4 activity may determine the timing of differentiation and delamination [42•]. Furthermore, Foxp4 gene expression was shown to be downstream of proneural gene activity, and knockdown of Foxp2/4 inhibited Neurog2-induced neuronal differentiation and migration out of the VZ [42•]. Experiments using mouse embryos suggested that a similar Foxp-dependent mechanism might also operate in the mammalian spinal cord and forebrain [42•,43]. N-cadherin-mediated adhesion can also be regulated by mechanisms other than transcriptional repression. In fact, Slit-Robo signaling, a well-known regulator of axon guidance, was recently found to inhibit N-cadherin mediated adhesion to promote apical process detachment [44•,45•]. This is consistent with the previous findings that the Slit-Robo pathway induces the phosphorylation of β-catenin and its

dissociation from N-cadherin, triggering dissolution of AJs [46]. However, it is unclear how Slit-Robo signaling interacts with neuronal fate commitment pathways. Taken together, these data suggest that modulation of N-cadherin-mediated adhesion can control the initiation of migration for newborn neurons or INPs. It will be interesting to test interactions between N-cadherin upstream regulators such as Foxp proteins and Slit-Robo signaling.

Another recent study has identified Scratch transcription factors as a molecular link between neuronal fate commitment and onset of radial migration in the developing mouse neocortex [47•]. Snail superfamily members Scratch1 and Scratch2 are expressed in newly fate committed neurons and INPs downstream of proneural genes Neurog1/2 and Ascl1 [47•]. In addition, knockdown and overexpression experiments demonstrated that Scratch1/2 regulate delamination from the ventricular surface through repression of E-cadherin [47•] (Figure 2b). Importantly, the Neurog2-induced onset of migration was inhibited by simultaneous knockdown of Scratch1/2 [47•], as is the case for Foxp2/4 [42•]. However, it is worth noting that, unlike Foxp2/4 [42•,43], Scratch1/2 appear to induce delamination of newborn neurons and INPs without enhancing neurogenesis [47•] (although they might have some effect on the neuronal/INP fate choice [48]), identifying separate pathways for differentiation and delamination. In fact, Scratch overexpression caused premature migration of Pax6-expressing cells out of the VZ [47•]. It is also interesting to note that the initiation of migration by the downregulation of cadherin-mediated adhesion that is described in this study mirrors neural crest cell migration in PNS development [49–51]. Furthermore, both here and in the PNS, Snail superfamily members are important in mediating this cadherin downregulation, which is characteristic of ephithelial–mesenchymal transitions (EMT) [47•,49–51]. An EMT is more generally characterized by the loss of epithelial features, such as intercellular adherens junctions, in exchange for mesenchymal features, such as high motility [49–51]. As these characteristics are present in neural crest cell and neuron/INP migration [47•,49–51], it seems that both the PNS and CNS use EMT-like mechanisms to trigger migration following cell fate specification.

The studies discussed here support the model that after division, when both daughters are attached to the apical surface, proneural pathways initiated in some cells not only activate neuronal fate commitment, but also promote detachment from the apical surface, allowing these cells to migrate basally (Figure 2b). Although the mechanisms for this coordination that are identified here do not obviously overlap with one another, it is reasonable to speculate that all these pathways are acting together to ensure the timely delamination of newborn neurons and INPs [39•,42•,47•]. One alternative explanation for the variety of pathways regulating neuronal migration onset is that there is diversity in neuronal subtypes between different CNS regions. The Foxp study largely focused on spinal cord [42•], whereas the Id and Scratch studies analyzed the developing neocortex [39•,47•]. Indeed, overexpression of Neurog2 in NPCs/RGs derived from mouse embryonic neocortex repressed E-cadherin (as in the Scratch pathway) but not N-cadherin (as in the Foxp pathway) [47•]. Distinct functions and/or different subcellular distributions of E-cadherin and N-cadherin may contribute to such regional differences [47•,52,53]. Besides transcriptional mechanisms, adhesion molecules are also regulated by several other mechanisms, including trafficking [29,54], that contribute to delamination. It is also possible that changes in centrosomal properties upon neuronal fate commitment may contribute to delamination. Newborn neurons and INPs preferentially inherit a daughter centriole, which generally exhibits less microtubule-organizing activity shortly after mitosis compared with the mother centriole inherited by RGs [55]. In addition, a recent study has reported an interesting observation that before delamination, newly fate committed cells harbor a primary cilia at the basolateral rather than apical position [56•]. While the significance of these differences between the centrosomes of newborn neurons/INPs and RGs is not clear, it is possible that centrosome regulation may act in concert and/or parallel with transcriptional mechanisms to trigger the disassembly of AJs and delamination. Further investigation is necessary for a comprehensive understanding of the mechanisms underlying the onset of migration of newborn neurons and INPs and to understand the pathogeneses of human diseases that may be caused by defects in neuronal migration onset [23,24,57–59].

Future directions

From the studies discussed here, it seems that delamination is triggered by the expression of proneural genes. However, precursors that are not attached to the ventricular surface, called outer RGs (oRGs, also known as basal RGs (bRGs)), were recently identified [60–62] (Figure 1b). These cells are believed to be produced from RGs, which raises the question of how they begin their migration away from the apical surface although they likely do not express proneural genes. It is possible that they have a different mechanism of detachment than (the majority of) neurons and INPs. For instance, a recent study using time-lapse imaging to observe oRG production proposed that they result from vertical spindle orientation divisions [63], although this finding contradicts an earlier study [61]. Studies eliminating LGN function, thus randomizing spindle orientation, support this idea: displaced progenitors in these experiments prematurely exit the ventricular zone and continue to self-renew [14,15,17•], thus resembling oRGs, not INPs [17•]. However, it is also possible that oRG detachment is regulated by the same molecules that regulate neuron and

INP detachment, but that these molecules are activated by different upstream factors in oRGs. Since Scratch proteins were shown to regulate delamination without affecting differentiation [47•], they could conceivably mediate oRG detachment. Because oRGs appear to be more common in gyrencephalic animals and are thought to be important for cortical expansion during evolution [64,65], further characterization of these cells will be important for understanding cortical evolution and human neurological disorders.

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References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- • of special interest

1. Gotz M, Huttner WB: **The cell biology of neurogenesis**. *Nat Rev Mol Cell Biol* 2005, **6**:777-788.

2. Kriegstein A, Alvarez-Buylla A: **The glial nature of embryonic and adult neural stem cells**. *Annu Rev Neurosci* 2009, **32**:149-184.

3. Guillemot F: **Spatial and temporal specification of neural fates by transcription factor codes**. *Development* 2007, **134**:3771-3780.

4. Ochiai W, Nakatani S, Takahara T, Kainuma M, Masaoka M, Minobe S, Namihira M, Nakashima K, Sakakibara A, Ogawa M *et al.*: **Periventricular notch activation and asymmetric Ngn2 and Tbr2 expression in pair-generated neocortical daughter cells**. *Mol Cell Neurosci* 2009, **40**:225-233.

5. Gupta A, Tsai LH, Wynshaw-Boris A: **Life is a journey: a genetic look at neocortical development**. *Nat Rev Genet* 2002, **3**:342-355.

6. Heng JI, Chariot A, Nguyen L: **Molecular layers underlying cytoskeletal remodelling during cortical development**. *Trends Neurosci* 2010, **33**:38-47.

7. Valiente M, Marin O: **Neuronal migration mechanisms in development and disease**. *Curr Opin Neurobiol* 2010, **20**:68-78.

8. Kawauchi T, Hoshino M: **Molecular pathways regulating cytoskeletal organization and morphological changes in migrating neurons**. *Dev Neurosci* 2008, **30**:36-46.

9. Chenn A, McConnell SK: **Cleavage orientation and the asymmetric inheritance of Notch1 immunoreactivity in mammalian neurogenesis**. *Cell* 1995, **82**:631-641.

10. Kosodo Y, Roper K, Haubensak W, Marzesco AM, Corbeil D, Huttner WB: **Asymmetric distribution of the apical plasma membrane during neurogenic divisions of mammalian neuroepithelial cells**. *EMBO J* 2004, **23**:2314-2324.

11. Sanada K, Tsai LH: **G protein betagamma subunits and AGS3 control spindle orientation and asymmetric cell fate of cerebral cortical progenitors**. *Cell* 2005, **122**:119-131.

12. Bultje RS, Castaneda-Castellanos DR, Jan LY, Jan YN, Kriegstein AR, Shi SH: **Mammalian Par3 regulates progenitor cell asymmetric division via notch signaling in the developing neocortex**. *Neuron* 2009, **63**:189-202.

13. Marthiens V, ffrench-Constant C: **Adherens junction domains are split by asymmetric division of embryonic neural stem cells**. *EMBO Rep* 2009, **10**:515-520.

14. Morin X, Jaouen F, Durbec P: **Control of planar divisions by the G-protein regulator LGN maintains progenitors in the chick neuroepithelium**. *Nat Neurosci* 2007, **10**:1440-1448.

15. Konno D, Shioi G, Shitamukai A, Mori A, Kiyonari H, Miyata T, Matsuzaki F: **Neuroepithelial progenitors undergo LGN-dependent planar divisions to maintain self-renewability during mammalian neurogenesis**. *Nat Cell Biol* 2008, **10**:93-101.

16. Noctor SC, Martinez-Cerdeno V, Kriegstein AR: **Distinct behaviors of neural stem and progenitor cells underlie cortical neurogenesis**. *J Comp Neurol* 2008, **508**:28-44.

17. • Shitamukai A, Konno D, Matsuzaki F: **Oblique radial glial divisions in the developing mouse neocortex induce self-renewing progenitors outside the germinal zone that resemble primate outer subventricular zone progenitors**. *J Neurosci* 2011, **31**:3683-3695.

This study shows that there is no correlation between the size of the apical surface inherited by a daugher cell and whether or not it delaminates from the ventricular surface. The authors also show that oblique or vetrical divisions can produce oRGs in the mouse cortex.

18. Morin X, Bellaiche Y: **Mitotic spindle orientation in asymmetric and symmetric cell divisions during animal development**. *Dev Cell* 2011, **21**:102-119.

19. • Postiglione MP, Juschke C, Xie Y, Haas GA, Charalambous C, Knoblich JA: **Mouse inscuteable induces apical-basal spindle orientation to facilitate intermediate progenitor generation in the developing neocortex**. *Neuron* 2011, **72**:269-284.

By altering the number of vertical spindle orientations in the mouse cortex via Inscuteable overexpression or ablation, this study describes how spindle orientation can influence the choice between INP and postmitotic neuron production.

20. Asami M, Pilz GA, Ninkovic J, Godinho L, Schroeder T, Huttner WB, Gotz M: **The role of Pax6 in regulating the orientation and mode of cell division of progenitors in the mouse cerebral cortex**. *Development* 2011, **138**:5067-5078.

21. Lancaster MA, Knoblich JA: **Spindle orientation in mammalian cerebral cortical development**. *Curr Opin Neurobiol* 2012, **22**:737-746.

22. Das RM, Storey KG: **Mitotic spindle orientation can direct cell fate and bias Notch activity in chick neural tube**. *EMBO Rep* 2012, **13**:448-454.

23. Fox JW, Lamperti ED, Eksioglu YZ, Hong SE, Feng Y, Graham DA, Scheffer IE, Dobyns WB, Hirsch BA, Radtke RA *et al.*: **Mutations in filamin 1 prevent migration of cerebral cortical neurons in human periventricular heterotopia**. *Neuron* 1998, **21**:1315-1325.

24. Sarkisian MR, Bartley CM, Rakic P: **Trouble making the first move: interpreting arrested neuronal migration in the cerebral cortex**. *Trends Neurosci* 2008, **31**:54-61.

25. Heng JI, Nguyen L, Castro DS, Zimmer C, Wildner H, Armant O, Skowronska-Krawczyk D, Bedogni F, Matter JM, Hevner R *et al.*: **Neurogenin 2 controls cortical neuron migration through regulation of Rnd2**. *Nature* 2008, **455**:114-118.

26. • Pacary E, Heng J, Azzarelli R, Riou P, Castro D, Lebel-Potter M, Parras C, Bell DM, Ridley AJ, Parsons M *et al.*: **Proneural transcription factors regulate different steps of cortical neuron migration through Rnd-mediated inhibition of RhoA signaling**. *Neuron* 2011, **69**:1069-1084.

This study shows that proneural bHLH factrors regulate the initial movement of newborn neuros and INPs through Rnd-dependent RhoA inhibition, linking neuronal fate commitment and initial neuron migration.

27. Kosodo Y, Suetsugu T, Suda M, Mimori-Kiyosue Y, Toida K, Baba SA, Kimura A, Matsuzaki F: **Regulation of interkinetic nuclear migration by cell cycle-coupled active and passive mechanisms in the developing brain**. *EMBO J* 2011, **30**:1690-1704.

28. Kadowaki M, Nakamura S, Machon O, Krauss S, Radice GL, Takeichi M: **N-cadherin mediates cortical organization in the mouse brain**. *Dev Biol* 2007, **304**:22-33.

29. Rasin MR, Gazula VR, Breunig JJ, Kwan KY, Johnson MB, Liu-Chen S, Li HS, Jan LY, Jan YN, Rakic P *et al.*: **Numb and Numbl are required for maintenance of cadherin-based adhesion and polarity of neural progenitors**. *Nat Neurosci* 2007, **10**:819-827.

30. Zhang J, Woodhead GJ, Swaminathan SK, Noles SR, McQuinn ER, Pisarek AJ, Stocker AM, Mutch CA, Funatsu N, Chenn A: **Cortical neural precursors inhibit their own differentiation via N-cadherin maintenance of beta-catenin signaling**. *Dev Cell* 2010, **18**:472-479.

31. Zechner D, Fujita Y, Hulsken J, Muller T, Walther I, Taketo MM, Crenshaw EB III, Birchmeier W: **Birchmeier C: beta-Catenin signals regulate cell growth and the balance between progenitor cell expansion and differentiation in the nervous system**. *Dev Biol* 2003, **258**:406-418.

32. Hand R, Bortone D, Mattar P, Nguyen L, Heng JI, Guerrier S, Boutt E, Peters E, Barnes AP, Parras C *et al.*: **Phosphorylation of Neurogenin2 specifies the migration properties and the dendritic morphology of pyramidal neurons in the neocortex**. *Neuron* 2005, **48**:45-62.

33. Ge W, He F, Kim KJ, Blanchi B, Coskun V, Nguyen L, Wu X, Zhao J, Heng JI, Martinowich K *et al.*: **Coupling of cell migration with neurogenesis by proneural bHLH factors**. *Proc Natl Acad Sci U S A* 2006, **103**:1319-1324.

34. Herzog D, Loetscher P, van Hengel J, Knusel S, Brakebusch C, Taylor V, Suter U, Relvas JB: **The small GTPase RhoA is required to maintain spinal cord neuroepithelium organization and the neural stem cell pool**. *J Neurosci* 2011, **31**:5120-5130.

35. Katayama K, Melendez J, Baumann JM, Leslie JR, Chauhan BK, Nemkul N, Lang RA, Kuan CY, Zheng Y, Yoshida Y: **Loss of RhoA in neural progenitor cells causes the disruption of adherens junctions and hyperproliferation**. *Proc Natl Acad Sci U S A* 2011, **108**:7607-7612.

36. Thumkeo D, Shinohara R, Watanabe K, Takebayashi H, Toyoda Y, Tohyama K, Ishizaki T, Furuyashiki T, Narumiya S: **Deficiency of mDia, an actin nucleator, disrupts integrity of neuroepithelium and causes periventricular dysplasia**. *PLoS ONE* 2011, **6**:e25465.

37. Cappello S, Bohringer CR, Bergami M, Conzelmann KK, Ghanem A, Tomassy GS, Arlotta P, Mainardi M, Allegra M, Caleo M *et al.*: **A radial glia-specific role of RhoA in double cortex formation**. *Neuron* 2012, **73**:911-924.

38. Pacary E, Azzarelli R, Guillemot F: **Rnd3 coordinates early steps of cortical neurogenesis through actin-dependent and -independent mechanisms**. *Nat Commun* 2013, **4**:1635.

39. • Niola F, Zhao X, Singh D, Castano A, Sullivan R, Lauria M, Nam HS, Zhuang Y, Benezra R, Di Bernardo D *et al.*: **Id proteins synchronize stemness and anchorage to the niche of neural stem cells**. *Nat Cell Biol* 2012, **14**:477-487.

This study shows that knockout of Id1–3 causes premature neurogenesis of NPCs concomitantly with their delamination from the ventricular surface in the developing cortex. Id1–3 contribute to apical attachment through suppression of Rap1GAP expression.

40. Kooistra MR, Dube N, Bos JL: **Rap1: a key regulator in cell–cell junction formation**. *J Cell Sci* 2007, **120**:17-22.

41. Loulier K, Lathia JD, Marthiens V, Relucio J, Mughal MR, Tang SC, Coksaygan T, Hall PE, Chigurupati S, Patton B *et al.*: **beta1 integrin maintains integrity of the embryonic neocortical stem cell niche**. *PLoS Biol* 2009, **7**:e1000176.

42. • Rousso DL, Pearson CA, Gaber ZB, Miquelajauregui A, Li S, Portera-Cailliau C, Morrisey EE, Novitch BG: **Foxp-mediated suppression of N-cadherin regulates neuroepithelial character and progenitor maintenance in the CNS**. *Neuron* 2012, **74**:314-330.

The authors show that, during motor neuron development in the chick spinal cord, Foxp2 and Foxp4 downregulate both N-cadherin and Sox2 to induce delamination and promote neuronal differentiation. Experiments using mouse embryos show that a similar Foxp-dependent mechanism might operate in the mammalian CNS.

43. Tsui D, Vessey JP, Tomita H, Kaplan DR, Miller FD: **FoxP2 regulates neurogenesis during embryonic cortical development**. *J Neurosci* 2013, **33**:244-258.

44. • Wong GK, Baudet ML, Norden C, Leung L, Harris WA: **Slit1b-Robo3 signaling and N-cadherin regulate apical process retraction in developing retinal ganglion cells**. *J Neurosci* 2012, **32**:223-228.

This paper shows that Slit1b-Robo3 signaling inhibits N-cadherin-mediated adhesion to promote apical process detachment in the developing zebrafish retina.

45. • Borrell V, Cardenas A, Ciceri G, Galceran J, Flames N, Pla R, Nobrega-Pereira S, Garcia-Frigola C, Peregrin S, Zhao Z *et al.*: **Slit/Robo signaling modulates the proliferation of central nervous system progenitors**. *Neuron* 2012, **76**:338-352.

This paper shows that Slit-Robo signaling promotes apical process detachment in the developing mouse cortex. This study also describes increased N-cadherin levels at the ventricular surface in the Robo1/2 mutant cortex.

46. Rhee J, Mahfooz NS, Arregui C, Lilien J, Balsamo J, VanBerkum MF: **Activation of the repulsive receptor Roundabout inhibits N-cadherin-mediated cell adhesion**. *Nat Cell Biol* 2002, **4**:798-805.

47. • Itoh Y, Moriyama Y, Hasegawa T, Endo TA, Toyoda T, Gotoh Y: **Scratch regulates neuronal migration onset via an epithelial–mesenchymal transition-like mechanism**. *Nat Neurosci* 2013, **16**:416-425.

This study shows that Snail superfamily members Scratch1/2 are expressed downstream of proneural bHLH factors and regulate delamination through repression of E-cadherin. The authors also describe the similarity between the Scratch-mediated neuronal migration onset and the typical EMT.

48. Paul V, Tonchev AB, Henningfeld KA, Pavlakis E, Rust B, Pieler T, Stoykova A: **Scratch2 modulates neurogenesis and cell migration through antagonism of bHLH proteins in the developing neocortex**. *Cereb Cortex* 2012. [Epub ahead of print].

49. Sauka-Spengler T, Bronner-Fraser M: **A gene regulatory network orchestrates neural crest formation**. *Nat Rev Mol Cell Biol* 2008, **9**:557-568.

50. Nieto MA: **The snail superfamily of zinc-finger transcription factors**. *Nat Rev Mol Cell Biol* 2002, **3**:155-166.

51. Thiery JP, Acloque H, Huang RY, Nieto MA: **Epithelial–mesenchymal transitions in development and disease**. *Cell* 2009, **139**:871-890.

52. Kan NG, Stemmler MP, Junghans D, Kanzler B, de Vries WN, Dominis M, Kemler R: **Gene replacement reveals a specific role for E-cadherin in the formation of a functional trophectoderm**. *Development* 2007, **134**:31-41.

53. Libusova L, Stemmler MP, Hierholzer A, Schwarz H, Kemler R: **N-cadherin can structurally substitute for E-cadherin during intestinal development but leads to polyp formation**. *Development* 2010, **137**:2297-2305.

54. Arvanitis DN, Behar A, Tryoen-Toth P, Bush JO, Jungas T, Vitale N, Davy A: **Ephrin B1 maintains apical adhesion of neural progenitors**. *Development* 2013, **140**:2082-2092.

55. Wang X, Tsai JW, Imai JH, Lian WN, Vallee RB, Shi SH: **Asymmetric centrosome inheritance maintains neural progenitors in the neocortex**. *Nature* 2009, **461**:947-955.

56. • Wilsch-Brauninger M, Peters J, Paridaen JT, Huttner WB: **Basolateral rather than apical primary cilia on neuroepithelial cells committed to delamination**. *Development* 2012, **139**:95-105.

This study demonstrates that newly fate-committed neurons or INPs harbor a primary cilia at the basolateral rather than apical position before delamination.

57. Sheen VL, Ganesh VS, Topcu M, Sebire G, Bodell A, Hill RS, Grant PE, Shugart YY, Imitola J, Khoury SJ *et al.*: **Mutations in ARFGEF2 implicate vesicle trafficking in neural progenitor proliferation and migration in the human cerebral cortex**. *Nat Genet* 2004, **36**:69-76.

58. Lai CS, Fisher SE, Hurst JA, Vargha-Khadem F, Monaco AP: **A forkhead-domain gene is mutated in a severe speech and language disorder**. *Nature* 2001, **413**:519-523.

59. O'Roak BJ, Vives L, Girirajan S, Karakoc E, Krumm N, Coe BP, Levy R, Ko A, Lee C, Smith JD *et al.*: **Sporadic autism exomes reveal a highly interconnected protein network of de novo mutations**. *Nature* 2012, **485**:246-250.

60. Hansen DV, Lui JH, Parker PR, Kriegstein AR: **Neurogenic radial glia in the outer subventricular zone of human neocortex**. *Nature* 2010, **464**:554-561.

61. Fietz SA, Kelava I, Vogt J, Wilsch-Brauninger M, Stenzel D, Fish JL, Corbeil D, Riehn A, Distler W, Nitsch R *et al.*: **OSVZ progenitors of human and ferret neocortex are epithelial-like and expand by integrin signaling**. *Nat Neurosci* 2010, **13**:690-699.

62. Wang X, Tsai JW, LaMonica B, Kriegstein AR: **A new subtype of progenitor cell in the mouse embryonic neocortex**. *Nat Neurosci* 2011, **14**:555-561.

63. Lamonica BE, Lui JH, Hansen DV, Kriegstein AR: **Mitotic spindle orientation predicts outer radial glial cell generation in human neocortex**. *Nat Commun* 2013, **4**:1665.

64. Fietz SA, Huttner WB: **Cortical progenitor expansion, self-renewal and neurogenesis-a polarized perspective**. *Curr Opin Neurobiol* 2011, **21**:23-35.

65. LaMonica BE, Lui JH, Wang X, Kriegstein AR: **OSVZ progenitors in the human cortex: an updated perspective on neurodevelopmental disease**. *Curr Opin Neurobiol* 2012, **22**:747-753.